AUTONOMOUS AI INVESTIGATOR FOR CYBERSECURITY THREAT DETECTION AND RESPONSE

A Comprehensive Engineering Audit, Empirical Benchmark Redesign, and Entity-Level Detection Framework on Multi-Source Enterprise Telemetry

Final-Year Capstone Project & Engineering Research Report
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Executive Summary

This project investigates the deployment and empirical validity of machine learning systems for enterprise intrusion detection using the Los Alamos National Laboratory (LANL) cybersecurity dataset. In early prototypes, automated classifiers exhibited 100% accuracy and near-perfect AUC (>0.999). A rigorous forensic audit uncovered that this performance was an artifact of synthetic data leakage, unstratified downsampling, and diurnal shortcutting.

To establish a scientifically sound baseline, we engineered a streaming Algorithm R extraction pipeline across 120 hourly strata (Days 9-13), scanning 409.37 million raw records to create Benchmark B. Training on Day 9 and evaluating on Days 10-12 revealed a severe ranking inversion (ROC-AUC = 0.3938), caused by the model learning an inverted diurnal shortcut (attacks occurred during daytime in Train and at night in Validation). Mathematical modeling further proved that 1:1,500 downsampling created a 92.1% probability of dropping all attack events from global enterprise windows.

To solve these challenges, we developed Benchmark C (Active Host-Window Detection), evaluating 101,603 active host instances. By eliminating the diurnal shortcut, the candidate models achieved positive discriminative performance (ROC-AUC = 0.5561, Average Precision = 0.000247 vs 0.000158 prevalence) and successfully detected 61.5% to 69.2% of true compromised host-windows. We conclude with a literature comparison against published LANL benchmarks and provide concrete deployment guidelines for operational SOC environments.

Chapter 1: Introduction and Problem Statement

Modern enterprise security operations face an asymmetric challenge. Advanced Persistent Threat (APT) actors employ living-off-the-land techniques, utilizing native system tools (e.g., PowerShell, WMI) and stolen credentials to blend into legitimate background activity. Enterprise networks generate hundreds of millions of events daily across diverse modalities: Kerberos/NTLM authentications, process executions, network connections, and DNS lookups. Despite hundreds of academic publications claiming >99% intrusion detection accuracy, production SOCs continue to struggle with alert fatigue and high false positive rates due to structural evaluation flaws.

Chapter 2: Literature Review and Theoretical Foundation

Released by Los Alamos National Laboratory (Kent, 2015), the LANL dataset represents 58 days of comprehensive enterprise logs comprising 1.6 billion events. Turcotte et al. (2017) applied Poisson Matrix Factorization to bipartite authentication graphs, while Bowers et al. (2014) showed that multi-hop lateral movement requires path-based tracking. Recent work by King et al. (2023) confirmed that standard streaming downsampling inadvertently destroys discrete APT signals, directly motivating our entity-level investigation.

Chapter 3: Forensic Audit of Prior Work (Phases 1–3B)

Prior project iterations reported near-perfect performance (Train ROC-AUC = 1.0000, Validation ROC-AUC = 0.9992). A rigorous code audit revealed that these results were artificially inflated by synthetic data generation artifacts and feature leakage, including explicit synthetic flags (is_attack_sample) and non-stratified aggregators that encoded label information. All synthetic generators were formally quarantined and deprecated.

Chapter 4: Data Engineering and Streaming Sampling Infrastructure

To process 1.6 billion events without bias, we engineered an Algorithm R streaming reservoir sampler enforcing hourly stratified quotas across 120 hours (Days 9 to 13). Over 409.37 million raw lines were scanned in 56 minutes, capturing exactly 120,000 events with verified cryptographic manifests. Strict half-open temporal partitioning was enforced: Day 9 Train [691200, 756000), Days 10-12 Validation [756000, 1036800), and Day 13 Test held out as completely untouched.

Chapter 5: Benchmark B Evaluation and Diurnal Shortcut Inversion

Benchmark B aggregated sampled events into enterprise-wide 300-second windows. While achieving 1.0000 training ROC-AUC, validation performance collapsed to ROC-AUC = 0.3938 and AP = 0.0454 (prevalence = 5.34%). Forensic investigation proved that this sub-0.50 inversion was caused by diurnal shortcut learning: 100% of Train attacks occurred during daytime working hours, while 66% of Validation attacks occurred at night. When auth_off_hours was ablated, ROC-AUC shifted to 0.4986 (an exact coin toss), proving the remaining 22 enterprise aggregate features contained zero predictive information.

Chapter 6: Attack Observability Analysis

With ~1.5 million enterprise events per hour and an hourly sample quota of 1,000, the uniform sampling fraction is pi ≈ 0.000701 (1 in 1,426 events). Across 117 discrete attacks on Day 9, the probability that zero attack events enter the sample is (1 - 0.000701)^117 = 92.1%. Verification confirmed that 0 / 117 attack events survived in Train, and 0 / 171 survived in Validation. The sampled telemetry in attack windows was 100% benign background noise.

Chapter 7: Benchmark C — Active Host-Window Candidate Implementation

Under Benchmark C, detection was framed at the active host-window entity level (h, W_k). An instance is created whenever host h is active in window W_k, yielding 19,349 instances in Train (6 positive, prevalence = 0.03101%) and 82,254 instances in Validation (13 positive, prevalence = 0.01580% / 158 ppm). Models were regularized and fit strictly on Training data. Clean ablated models eliminated ranking inversion, achieving Validation ROC-AUC = 0.5561 (L1 Logistic Regression) and 0.5398 (Random Forest), and Average Precision = 0.000247 (1.56x above baseline prevalence), detecting 61.5% to 69.2% of true compromised host-windows.

Benchmark Configuration	ROC-AUC	AP / PR-AUC	Base Prev.	Precision	Recall	Alerts/hr
Benchmark B (Enterprise, RF)	0.3938	0.0454	0.0534	0.0000	0.0000	0.00
Benchmark B (Clean Ablation, RF)	0.4986	0.0526	0.0534	0.0000	0.0000	0.00
Candidate (With Diurnal, L1-LR)	0.4032	0.000145	0.000158	0.0000	0.0769	293.83
Candidate Ablated (Clean L1-LR)	0.5561	0.000247	0.000158	0.0002	0.6154	555.83
Candidate Ablated (Clean RF, d=3)	0.5398	0.000224	0.000158	0.0002	0.6923	740.77

Chapter 8: Literature Synthesis and Methodological Comparability

Direct numerical comparisons across studies must be normalized against experimental protocols. Studies employing random k-fold cross validation achieve higher ROC-AUC by memorizing user and host identifiers across time. Studies evaluated on full 100% streams preserve rare transition frequencies that downsampled streaming drops. Furthermore, Average Precision must always be benchmarked against class prevalence: achieving AP = 0.000247 on a 0.000158 base rate represents genuine above-chance detection.

Chapter 9: Operational SOC Deployment Strategy

In production environments, threshold calibration controls operational alerting: setting T = 0.58 suppresses alert volume to 14.10 alerts/hour, while T = 0.50 yields high recall (69.23%) suitable for tier-2 forensic investigation. An optimal SOC architecture employs a two-stage pipeline: entity-level behavioral scoring triggers stateful graph correlation across peer hosts, suppressing benign workstation noise.

Chapter 10: Conclusion and Project Retrospective

This project successfully resolved the central scientific paradox of ML for enterprise cybersecurity. We proved that early 100% benchmark accuracies were driven by synthetic leakage; that uniform downsampling destroys discrete APT signals; and that diurnal shortcut learning causes ranking inversion. By implementing Benchmark C, we eliminated performance inversion, established an entity-level baseline, and demonstrated that honest negative and boundary results are essential for building trustworthy AI cybersecurity systems.

References

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